

20260322 Setonix Shared Set up

Due to pawsey policy, we have to use containers to avoid installing large numbers of small files (common with Conda/Mamba) directly.

Below is a **clean, corrected project-wide installation guide** for reinstalling the shared stack under:

```
/software/projects/pawsey1308
```

Clarifications:

- shared/common things go directly under
`/software/projects/pawsey1308`
 - private CADC credentials should live under each user's own existing folder, e.g.
 - `/software/projects/pawsey1308/$USER/cadc_home`
 - shared wrapper commands should live in:
 - `/software/projects/pawsey1308/bin`
-

Project-wide shared software setup for pawsey1308

0. Final layout

After setup, the shared/common structure should look like this:

```
/software/projects/pawsey1308/
├─ bin/
│   ├── conda1308
│   ├── mamba1308
│   ├── ngistenv1308
│   ├── mapviewer1308
│   ├── cadc-get-cert
│   ├── vcp
│   └─ vls
├─ containers/
│   ├── mambaforge_latest.sif
│   ├── ngistenv1308_overlay.img
│   └─ cadc_overlay.img
├─ docs/
├─ ngist/
├─ ngist_supplementary_public/
├─ setonix/                # optional helper scripts / slurm
templates
├─ rhuang/                 # personal folder, already exists
```

Meaning of each area

- **/software/projects/pawsey1308/ngist**
shared nGIST repo
- **/software/projects/pawsey1308/ngist_supplementary_public**
shared supplementary repo
- **/software/projects/pawsey1308/containers**
shared container image and shared overlays
- **/software/projects/pawsey1308/bin**

shared wrapper commands for everyone

- `/software/projects/pawsey1308/$USER/cadc_home`
each user's private CADc certificate store
 - `/scratch/pawsey1308/$USER/...`
per-user caches, logs, personal softwares
-

Part A — Maintainer installation steps (For Rongjun only)

This section is for the person setting up the shared project-wide stack.

1. Create the shared directories

Run:

```
BASE=/software/projects/pawsey1308

mkdir -p $BASE/bin
mkdir -p $BASE/containers
mkdir -p $BASE/docs
mkdir -p $BASE/setonix
```

2. Clone the shared repositories

2.1 nGIST repo

Choose whether the project should track `main`, `dev-branch`, or a specific tag. For a shared install, I recommend using `main` unless the project explicitly wants `dev-branch`.

```
cd /software/projects/pawsey1308
git clone https://github.com/geckos-survey/ngist.git
cd ngist
git checkout main
git pull --ff-only origin main
```

2.2 supplementary repo

```
cd /software/projects/pawsey1308
git clone https://github.com/geckos-
survey/ngist_supplementary_public.git
cd ngist_supplementary_public
git checkout main
git pull --ff-only origin main
```

3. Pull the shared base container

We will use one shared base Singularity image for both the nGIST environment and the CADIC tools.

```
module load singularity/4.1.0-slurm

cd /software/projects/pawsey1308/containers
singularity pull docker://condaforge/mambaforge:latest
```

This should create:

```
/software/projects/pawsey1308/containers/mambaforge_latest.sif
```

4. Create the shared nGIST overlay

This overlay will contain the shared environment `/ngistenv1308`.

```

BASE=/software/projects/pawsey1308
SCR=/scratch/pawsey1308/$USER
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/ngistenv1308_overlay.img

mkdir -p $SCR/fake_home_1308
mkdir -p $SCR/pip_cache_1308
mkdir -p $SCR/conda_pkgs_1308

cd $BASE/containers
singularity overlay create --size 51200 "$OVERLAY"

```

The overlay is used because the environment contains lots of small files, and we do not want them to count as normal files on `/software`.

5. Build the shared nGIST environment inside the overlay

Install the environment as:

```
/ngistenv1308
```

inside the overlay.

```

module load singularity/4.1.0-slurm

BASE=/software/projects/pawsey1308
SCR=/scratch/pawsey1308/$USER
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/ngistenv1308_overlay.img

singularity exec --cleanenv \
  --bind $BASE:/work \
  --bind $SCR:$SCR \
  --overlay $OVERLAY \
  $IMG \
  bash -lc "
    export HOME=$SCR/fake_home_1308
    export XDG_CACHE_HOME=$SCR/pip_cache_1308

```

```

export PIP_CACHE_DIR=$SCR/pip_cache_1308
export CONDA_PKGS_DIRS=$SCR/conda_pkgs_1308
export PYTHONNOUSERSITE=1

source /opt/conda/etc/profile.d/conda.sh

mamba env create -f
/work/ngist_supplementary_public/ngistenv.yaml -p
/ngistenv1308
  /ngistenv1308/bin/python -m pip install -e /work/ngist --
no-deps
"

```

Test the environment

```

module load singularity/4.1.0-slurm

BASE=/software/projects/pawsey1308
SCR=/scratch/pawsey1308/$USER
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/ngistenv1308_overlay.img

singularity exec --cleanenv \
  --bind $BASE:/work \
  --bind $SCR:$SCR \
  --overlay $OVERLAY \
  $IMG \
  bash -lc "
    /ngistenv1308/bin/python -V
    /ngistenv1308/bin/python -m pip show ngistPipeline | egrep
'Name|Version|Location'
    /ngistenv1308/bin/ngistPipeline --help | head
"

```

6. Create the shared CADC overlay

This overlay contains the CADC/VOS tools, but **not** shared credentials.

```
BASE=/software/projects/pawsey1308
SCR=/scratch/pawsey1308/$USER
IMG=$BASE/containers/mambaforge_latest.sif
CADC_OVERLAY=$BASE/containers/cadc_overlay.img

mkdir -p $SCR/fake_home_cadc
mkdir -p $SCR/pip_cache_cadc
mkdir -p $SCR/conda_pkgs_cadc

cd $BASE/containers
singularity overlay create --size 4096 "$CADC_OVERLAY"
```

7. Install CADC/VOS tools into the CADC overlay

Install them into `/cadcenv` inside the overlay.

```
module load singularity/4.1.0-slurm

BASE=/software/projects/pawsey1308
SCR=/scratch/pawsey1308/$USER
IMG=$BASE/containers/mambaforge_latest.sif
CADC_OVERLAY=$BASE/containers/cadc_overlay.img

singularity exec --cleanenv \
  --bind $BASE:/work \
  --bind $SCR:$SCR \
  --overlay $CADC_OVERLAY \
  $IMG \
  bash -lc "
    export HOME=$SCR/fake_home_cadc
    export XDG_CACHE_HOME=$SCR/pip_cache_cadc
    export PIP_CACHE_DIR=$SCR/pip_cache_cadc
    export CONDA_PKGS_DIRS=$SCR/conda_pkgs_cadc
    export PYTHONNOUSERSITE=1

    source /opt/conda/etc/profile.d/conda.sh

    mamba create -y -p /cadcenv python=3.11 pip
```



```
conda run -p /cadcenv pip install --no-cache-dir cadcutils  
vos  
"
```

Part B — Create the shared wrapper commands (For Rongjun only)

All of the following scripts go into:

```
/software/projects/pawsey1308/bin
```

These are the commands that everyone in the project will use.

1. Shared `conda1308`

Create:

```
cat > /software/projects/pawsey1308/bin/conda1308 <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

module load singularity/4.1.0-slurm >/dev/null 2>&1 || true

BASE=/software/projects/pawsey1308
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/ngistenv1308_overlay.img

SCR=/scratch/pawsey1308/${USER}
FAKE_HOME=$SCR/fake_home_1308
CACHE=$SCR/conda_cache_1308
TMP=$SCR/singularity_tmp_1308
PKGS=$SCR/conda_pkgs_1308

mkdir -p "$FAKE_HOME" "$CACHE" "$TMP" "$PKGS"

export SINGULARITY_TMPDIR="$TMP"
export SINGULARITY_CACHEDIR="$CACHE"

exec singularity exec --cleanenv \
    --bind "$BASE:/work" \
```

```

--bind "$SCR:$SCR" \
--overlay "$OVERLAY" \
"$IMG" \
bash -lc "
    export HOME=$FAKE_HOME
    export XDG_CACHE_HOME=$CACHE
    export PIP_CACHE_DIR=$CACHE
    export CONDA_PKGS_DIRS=$PKGS
    export PYTHONNOUSERSITE=1
    source /opt/conda/etc/profile.d/conda.sh
    conda \"\$@\"
" bash "$@"
EOF

chmod 755 /software/projects/pawsey1308/bin/conda1308

```

2. Shared `mamba1308`

```

cat > /software/projects/pawsey1308/bin/mamba1308 <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

module load singularity/4.1.0-slurm >/dev/null 2>&1 || true

BASE=/software/projects/pawsey1308
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/ngistenv1308_overlay.img

SCR=/scratch/pawsey1308/${USER}
FAKE_HOME=$SCR/fake_home_1308
CACHE=$SCR/conda_cache_1308
TMP=$SCR/singularity_tmp_1308
PKGS=$SCR/conda_pkgs_1308

mkdir -p "$FAKE_HOME" "$CACHE" "$TMP" "$PKGS"

export SINGULARITY_TMPDIR="$TMP"
export SINGULARITY_CACHEDIR="$CACHE"

```

```

exec singularity exec --cleanenv \
  --bind "$BASE:/work" \
  --bind "$SCR:$SCR" \
  --overlay "$OVERLAY" \
  "$IMG" \
  bash -lc "
    export HOME=$FAKE_HOME
    export XDG_CACHE_HOME=$CACHE
    export PIP_CACHE_DIR=$CACHE
    export CONDA_PKGS_DIRS=$PKGS
    export PYTHONNOUSERSITE=1
    source /opt/conda/etc/profile.d/conda.sh
    mamba "\"\${@}\""
    " bash "$@"
EOF

chmod 755 /software/projects/pawsey1308/bin/mamba1308

```

3. Shared ngistenv1308

```

cat > /software/projects/pawsey1308/bin/ngistenv1308 <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
exec /software/projects/pawsey1308/bin/conda1308 run -p
/ngistenv1308 "$@"
EOF

chmod 755 /software/projects/pawsey1308/bin/ngistenv1308

```

Usage example later will be:

```

ngistenv1308 python -V
ngistenv1308 ngistPipeline --help

```

4. Shared `mapviewer1308`

```
cat > /software/projects/pawsey1308/bin/mapviewer1308 <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

module load singularity/4.1.0-slurm >/dev/null 2>&1 || true

BASE=/software/projects/pawsey1308
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/ngistenv1308_overlay.img

export SINGULARITYENV_DISPLAY="${DISPLAY:-}"
export SINGULARITYENV_XAUTHORITY="$HOME/.Xauthority"
export SINGULARITYENV_QT_X11_NO_MITSHM=1

exec singularity exec --cleanenv \
  --bind /tmp/.X11-unix:/tmp/.X11-unix \
  --bind "$HOME/.Xauthority:$HOME/.Xauthority" \
  --overlay "$OVERLAY" \
  "$IMG" \
  /ngistenv1308/bin/Mapviewer "$@"
EOF

chmod 755 /software/projects/pawsey1308/bin/mapviewer1308
```

5. Shared `cadc-get-cert`

Each user's CADC certificate is stored in:

```
/software/projects/pawsey1308/$USER/cadc_home
```

Create:

```
cat > /software/projects/pawsey1308/bin/cadc-get-cert <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
```

```

module load singularity/4.1.0-slurm >/dev/null 2>&1 || true

BASE=/software/projects/pawsey1308
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/cadc_overlay.img

SCR=/scratch/pawsey1308/${USER}
CADC_HOME=$BASE/${USER}/cadc_home

mkdir -p "$SCR/singularity_tmp_cadc"
"$SCR/singularity_cache_cadc" "$CADC_HOME"
chmod 700 "$CADC_HOME" 2>/dev/null || true

export SINGULARITY_TMPDIR="$SCR/singularity_tmp_cadc"
export SINGULARITY_CACHEDIR="$SCR/singularity_cache_cadc"

exec singularity exec --cleanenv \
  --bind "$BASE:/work" \
  --bind "$SCR:$SCR" \
  --overlay "$OVERLAY" \
  "$IMG" \
  bash -lc '
    export HOME=/work/"${USER}" /cadc_home
    /cadcenv/bin/cadc-get-cert "$@"
    ' bash "$@"
EOF

chmod 755 /software/projects/pawsey1308/bin/cadc-get-cert

```

6. Shared vcp

```

cat > /software/projects/pawsey1308/bin/vcp <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

module load singularity/4.1.0-slurm >/dev/null 2>&1 || true

```

```

BASE=/software/projects/pawsey1308
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/cadc_overlay.img

SCR=/scratch/pawsey1308/${USER}
CADC_HOME=$BASE/${USER}/cadc_home

mkdir -p "$SCR/singularity_tmp_cadc"
"$SCR/singularity_cache_cadc" "$CADC_HOME"
chmod 700 "$CADC_HOME" 2>/dev/null || true

export SINGULARITY_TMPDIR="$SCR/singularity_tmp_cadc"
export SINGULARITY_CACHEDIR="$SCR/singularity_cache_cadc"

exec singularity exec --cleanenv \
  --bind "$BASE:/work" \
  --bind "$SCR:$SCR" \
  --overlay "$OVERLAY" \
  "$IMG" \
  bash -lc '
    export HOME=/work/"${USER}" /cadc_home
    /cadcenv/bin/vcp "$@"
    ' bash "$@"
EOF

chmod 755 /software/projects/pawsey1308/bin/vcp

```

7. Shared vls

```

cat > /software/projects/pawsey1308/bin/vls <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

module load singularity/4.1.0-slurm >/dev/null 2>&1 || true

BASE=/software/projects/pawsey1308
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/cadc_overlay.img

```

```
SCR=/scratch/pawsey1308/${USER}
CADC_HOME=$BASE/${USER}/cadc_home

mkdir -p "$SCR/singularity_tmp_cadc"
"$SCR/singularity_cache_cadc" "$CADC_HOME"
chmod 700 "$CADC_HOME" 2>/dev/null || true

export SINGULARITY_TMPDIR="$SCR/singularity_tmp_cadc"
export SINGULARITY_CACHEDIR="$SCR/singularity_cache_cadc"

exec singularity exec --cleanenv \
  --bind "$BASE:/work" \
  --bind "$SCR:$SCR" \
  --overlay "$OVERLAY" \
  "$IMG" \
  bash -lc '
    export HOME=/work/'"${USER}"'/cadc_home
    /cadcenv/bin/vls "$@"
  ' bash "$@"
EOF

chmod 755 /software/projects/pawsey1308/bin/vls
```

Part C — what every team member needs to do (For Everyone)

This is the user-side setup.

1. Add the shared project `bin` to `PATH`

Each user adds this line to their own `~/.bashrc`:

```
export PATH=/software/projects/pawsey1308/bin:$PATH
```

Then reload:

```
source ~/.bashrc
```

That is enough.

Users do **not** need to copy these scripts into `~/bin`.

2. Check that the shared stack works

Run:

```
conda1308 --version
mamba1308 --version
ngistenv1308 python -V
ngistenv1308 ngistPipeline --help | head
```

Part D — how everyone should use the shared stack (For Everyone)

1. Using nGIST from the command line

Basic usage

```
ngistenvl308 ngistPipeline --config  
configFiles/MasterConfig.yaml --default-dir  
configFiles/defaultDir
```

Running Python inside the shared environment

```
ngistenvl308 python -V  
ngistenvl308 python -c "import ngistPipeline; print('ok')"
```

2. Using Mapviewer

Users must connect with X forwarding:

```
ssh -Y <username>@setonix-03.pawsey.org.au
```

Then:

```
mapviewer1308
```

If they are on macOS, XQuartz must be running locally.

3. Using CADc tools

First-time setup for each user

Each user must obtain their own CADC proxy certificate:

```
cadc-get-cert -u <their_CADC_username>
```

This stores the credential under:

```
/software/projects/pawsey1308/$USER/cadc_home/.ssl/cadcproxy.pem
```

List ARC files

```
vls arc:projects/mauve/cubes/v3tk
```

Copy an ARC file to scratch

```
mkdir -p /scratch/pawsey1308/$USER/mauve/cubes/v3tk

vcp \

arc:projects/mauve/cubes/v3tk/IC3392_DATACUBE_FINAL_WCS_Pall_m
ad_red_v3tk.fits.gz \
    /scratch/pawsey1308/$USER/mauve/cubes/v3tk/
```

4. Using the shared nGIST stack in Slurm jobs

Use the shared wrapper directly.

Example job script under

```
/software/projects/pawsey1308/ngist_supplementary_public/ngistTutorial
1 using the shared environment:
```

```
#!/bin/bash -l

#SBATCH --account=pawsey1308
#SBATCH --partition=work
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=128
#SBATCH --mem=220G
#SBATCH --time=24:00:00
#SBATCH --job-name=IC3392_v3tk_v7.6.8
#SBATCH --output=slurm-%j.out
#SBATCH --error=slurm-%j.err
#SBATCH --mail-
user=rongjun.huang@research.uwa.edu.au,astro@rongjun-huang.com
#SBATCH --mail-type=BEGIN,END,FAIL

set -euo pipefail

module load singularity/4.1.0-slurm

cd
/software/projects/pawsey1308/ngist_supplementary_public/ngist
Tutorial

# (Optional but usually wise) prevent oversubscription from
BLAS/OpenMP libs:
export OMP_NUM_THREADS=1
export OPENBLAS_NUM_THREADS=1
export MKL_NUM_THREADS=1

# run + stream output to both screen (slurm out) and logfile
srun -n 1 -c $SLURM_CPUS_PER_TASK stdbuf -oL -eL \
  ngistenv1308 ngistPipeline \
  --config
configFiles/IC3392_MAUVE_MasterConfig_v7.6.8_setonix.yaml \
  --default-dir configFiles/defaultDir \
  2>&1 | tee IC3392_v3tk_v7.6.8.log

exit ${PIPESTATUS[0]}
```

Part E — update procedure for the maintainer (For Rongjun only)

When the shared stack needs updating:

1. Update the repos

```
cd /software/projects/pawsey1308/ngist
git fetch --all --tags
git checkout main
git pull --ff-only

cd /software/projects/pawsey1308/ngist_supplementary_public
git fetch --all --tags
git checkout main
git pull --ff-only
```

2. Update the shared environment

```
module load singularity/4.1.0-slurm

BASE=/software/projects/pawsey1308
SCR=/scratch/pawsey1308/$USER
IMG=$BASE/containers/mambaforge_latest.sif
OVERLAY=$BASE/containers/ngistenv1308_overlay.img

mkdir -p $SCR/fake_home_1308
mkdir -p $SCR/pip_cache_1308
mkdir -p $SCR/conda_pkgs_1308

singularity exec --cleanenv \
  --bind $BASE:/work \
  --bind $SCR:$SCR \
  --overlay $OVERLAY \
  $IMG \
```

```
bash -lc "  
  export HOME=$SCR/fake_home_1308  
  export XDG_CACHE_HOME=$SCR/pip_cache_1308  
  export PIP_CACHE_DIR=$SCR/pip_cache_1308  
  export CONDA_PKGS_DIRS=$SCR/conda_pkgs_1308  
  export PYTHONNOUSERSITE=1  
  source /opt/conda/etc/profile.d/conda.sh  
  
  mamba env update -p /ngistenv1308 -f  
/work/ngist_supplementary_public/ngistenv.yaml  
  /ngistenv1308/bin/python -m pip install -e /work/ngist --  
no-deps  
"
```

Re-test

```
ngistenv1308 python -m pip show ngistPipeline | egrep  
'Name|Version|Location'  
ngistenv1308 ngistPipeline --help | head
```

Final operational summary

Shared/common

- `/software/projects/pawsey1308/bin`
- `/software/projects/pawsey1308/containers`
- `/software/projects/pawsey1308/ngist`
- `/software/projects/pawsey1308/ngist_supplementary_public`
- `/software/projects/pawsey1308/docs`
- `/software/projects/pawsey1308/setonix`

Per-user private

- `/software/projects/pawsey1308/$USER/cadc_home`

Command names for the team

- `conda1308`
- `mamba1308`
- `ngistenv1308`
- `mapviewer1308`
- `cadc-get-cert`
- `vcp`
- `vls`